

● MEDIUM

● ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

04

10 questions · ~15 min

Q1

ADVANCED MATH

$$(x+2)(x+3) = (x-2)(x-3) + 10$$

Which of the following is a solution to the given equation?

- A. 1
- B. 0
- C. -2
- D. -5

$$y = x^2$$

$$2y + 6 = 2(x + 3)$$

If (x, y) is a solution of the system of equations above and $x > 0$, what is the value of xy ?

- A. 1
- B. 2
- C. 3
- D. 9

Q3

GRID-IN

ADVANCED MATH

$$x^2 + x - 12 = 0$$

If a is a solution of the equation above and $a > 0$, what is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ADVANCED MATH

$$y = x - 2x + 4$$

$$y = 6x - 12$$

Which ordered pair x, y is the solution to the given system of equations?

- A. 0, 2
- B. -4, 2
- C. 2, 0
- D. 2, -4

Q5

ADVANCED MATH

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A. $x = \frac{7by}{11}$
- B. $x = y - 77b$
- C. $x = \frac{y}{77b}$
- D. $x = 77by$

$$(x-4)(x+2)(x-1)=0$$

What is the product of the solutions to the given equation?

- A. 8
- B. 3
- C. -3
- D. -8

$$6x^2 + 5x - 7 = 0$$

What are the solutions to the given equation?

A. $\frac{-5 \pm \sqrt{25 + 168}}{12}$

B. $\frac{-6 \pm \sqrt{25 + 168}}{12}$

C. $\frac{-5 \pm \sqrt{36 - 168}}{12}$

D. $\frac{-6 \pm \sqrt{36 - 168}}{12}$

Q8

ADVANCED MATH

$$p = \frac{k}{4j + 9}$$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

- A. $4j + 9 = \frac{k}{p}$
- B. $4j + 9 = kp$
- C. $4j + 9 = k - p$
- D. $4j + 9 = \frac{p}{k}$

Q9

ADVANCED MATH

The equation $12t + b = c$ relates the variables t , b , and c . Which of the following correctly expresses the value of $c - b$ in terms of t ?

- A. $\frac{t}{12}$
- B. t
- C. $t + \frac{1}{12}$
- D. $12t$

$$w^2 + 12w - 40 = 0$$

Which of the following is a solution to the given equation?

- A. $6 - 2\sqrt{19}$
- B. $2\sqrt{19}$
- C. $\sqrt{19}$
- D. $-6 + 2\sqrt{19}$